

Adaptive Kernel Selection for Stein Variational Gradient Descent

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Abstract

Efficiently approximating intractable target distributions is a pervasive challenge across Bayesian inference and, more broadly, across probabilistic machine learning. Stein Variational Gradient Descent (SVGD) is a widely adopted variational inference method that transports a set of interacting particles so as to approximate an arbitrary target distribution. Because the SVGD dynamics are governed by a reproducing kernel Hilbert space (RKHS), they are highly sensitive to the choice of kernel function, a choice that directly shapes both convergence behavior and the quality of the resulting approximation across problem settings. The commonly used median heuristic offers a simple recipe for setting kernel bandwidths, but its rigidity limits its reach and it frequently performs poorly, especially as problem dimensionality grows. In this work, we develop a general strategy for adaptively choosing kernel parameters over an essentially arbitrary parameterized family of kernels. Recent convergence analyses based on the kernelized Stein discrepancy (KSD) suggest that tuning the kernel parameters by maximizing the KSD can improve performance, and this principle applies broadly rather than to any single kernel form. Building on this insight, we introduce Adaptive SVGD (Ad-SVGD), a method that alternates between transporting the particles via SVGD and adaptively tuning the kernel parameters through gradient ascent on the KSD. We provide a streamlined theoretical analysis that lifts existing fixed-kernel KSD-minimization results to our adaptive setting, establishing convergence properties for the maximal KSD taken over the entire kernel class. Our empirical results reinforce this picture: across a diverse collection of inference tasks and dimensionalities, Ad-SVGD consistently outperforms standard heuristics.

1 Introduction

Stein Variational Gradient Descent (SVGD) [18] is a deterministic, particle-based method for approximate inference that has emerged as a versatile alternative to traditional Markov Chain Monte Carlo (MCMC) methods across a wide range of probabilistic modeling problems. SVGD evolves a set of interacting particles using update directions derived from the functional gradient of the Kullback-Leibler (KL) divergence to the target distribution, with updates constrained to the unit ball of a reproducing kernel Hilbert space (RKHS). This general recipe applies to essentially any differentiable target density and is agnostic to the specific application domain. A critical and recurring limitation of SVGD, observed across problem settings, is its sensitivity to kernel choice, which strongly governs the algorithm’s overall performance [8, 28]. In addition, the resulting particle approximation tends to underestimate the posterior variance [2]. Together, these observations have fostered a widely held belief that SVGD generally fails to scale as the dimension of the underlying state space grows. In this work, we challenge this belief in broad terms by introducing an adaptive mechanism for kernel parameter selection that dynamically tunes the kernel during inference by maximizing the kernelized Stein discrepancy (KSD), enabling more effective transport across complex and high-dimensional spaces and across diverse target geometries.

1.1 Related work.

Since its introduction [18], SVGD has grown into a broadly applicable tool for approximate Bayesian inference, finding use across a diverse array of machine learning applications [22, 24, 30, 15]. In parallel, a substantial body of work has advanced the theoretical understanding of the method. Mean-field convergence has been analyzed in both continuous-time [23, 8, 5] and discrete-time [16, 32] settings, and finite-particle convergence rates have been established under a variety of assumptions [3, 33]. More broadly, the SVGD dynamics have been connected to gradient flows on spaces of probability distributions [19, 8], in analogy with Jordan, Kinderlehrer, and Otto [14], situating the method within a general geometric framework.

Across all of these settings, the performance of SVGD hinges critically on the choice of the kernel function, since the kernel governs the interaction between particles and the overall convergence of the method (see also Figures 1 and 9). Convergence results generally refer to mean-field convergence with respect to KSD, whose relation to weak convergence itself depends on the selected kernel [11]. The widely used median heuristic [12] offers a simple implementation but lacks theoretical justification and is known to degrade in performance across problems as their dimensionality increases [2, 40, 36]. A range of subsequent work has developed tools to mitigate this degradation in high-dimensional settings [7, 37, 10, 21]. The algorithm has also been extended through neural networks that learn the update direction [17, 39].

The approach most closely related to our work is Ai et al. [1], which introduces a mixture-of-kernels framework. Their method defines a KSD for a weighted linear combination of kernels and learns the kernel weights by maximizing this multiple-kernel KSD. Their construction, however, is confined to finite kernel bases and does not address the continuous, general parameter optimization over arbitrary kernel families that we develop here.

1.2 Contributions.

In this work, we address the fundamental and broadly relevant research question of whether SVGD, without any architectural or algorithmic enhancements, can achieve competitive performance across inference problems when equipped with a principled, adaptively chosen kernel. By deliberately isolating kernel selection from the many other modifications found in SVGD variants, we aim to understand, in general terms, the extent to which kernel choice alone governs the effectiveness of SVGD and how far adaptive kernel learning can push the capabilities of the original method across a spectrum of tasks.

Our main contributions are as follows.

- (i) **Adaptive Kernel Selection Method.** We propose a general method that dynamically updates the kernel parameters by maximizing the KSD during SVGD inference. In contrast to the commonly used median heuristic, which relies on a single scalar bandwidth, our approach optimizes an arbitrary number of continuous kernel parameters over essentially any parameterized kernel family, enabling far greater flexibility and adaptivity during SVGD updates.
- (ii) **Theoretical Analysis.** We provide theoretical motivation by analyzing our algorithm in the discrete-time mean-field setting, extending existing convergence results for SVGD with fixed kernels to the adaptive case. Specifically, we show that the supremum of the KSD over an arbitrary parameterized kernel class converges to zero as the particle distribution

approaches the target. Assuming a Stein logarithmic Sobolev inequality, we further derive iteration complexity in the mean-field limit.

- (iii) **Empirical Validation.** Through numerical experiments spanning a range of targets and dimensionalities, we demonstrate that our adaptive kernel selection consistently outperforms the median heuristic and helps alleviate variance collapse.

2 Mathematical background

We begin by considering a symmetric positive definite kernel $k : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ and its associated RKHS $\mathcal{H}_0$. We define $\mathcal{H}$ as the d -fold Cartesian product of $\mathcal{H}_0$ equipped with the inner product $\langle f, g \rangle_{\mathcal{H}} = \sum_{i=1}^d \langle f_i, g_i \rangle_{\mathcal{H}_0}$ and the canonical feature map $\Phi_k(x) = k(\cdot, x) \in \mathcal{H}_0$. Moreover, we denote by $x \cdot y$ the Euclidean inner product and $\nabla \cdot$ the divergence operator. The space of probability measures on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ is denoted by $\mathcal{P}(\mathbb{R}^d)$ and $\mathcal{P}_p(\mathbb{R}^d)$ denotes the subspace of measures with finite p -th moment. For $\mu, \nu \in \mathcal{P}_p(\mathbb{R}^d)$, we define the Wasserstein p -distance

$$\mathcal{W}_p(\mu, \nu) = \inf_{\gamma \in \Gamma(\mu, \nu)} \left(\int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|_2^p d\gamma(x, y) \right)^{1/p},$$

where $\Gamma(\mu, \nu)$ is the set of couplings of μ and ν , i.e. the set of probability measures on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals μ and ν .

2.1 Integral probability metrics and kernelized Stein discrepancy.

Integral probability metrics (IPMs) [26] constitute a general family of distances between measures, quantifying the maximum deviation of integrals over a chosen class of test functions. To render this approach tractable when measuring the distance of a sample to an intractable target, Stein’s method [35] provides a principled way to construct test functions with zero mean with respect to the target, applicable to any sufficiently regular target density. Indeed, for the operator $\mathcal{S}_\pi f := \nabla \log \pi \cdot f + \nabla \cdot f$ and a suitable choice of kernel, we have $\int_{\mathbb{R}^d} \mathcal{S}_\pi f(x) d\pi(x) = 0$ for all $f \in \mathcal{H}$ [6, 20]. This construction gives rise to the *kernelized Stein discrepancy* (KSD),

$$\text{KSD}(\mu|\pi) := \sup_{f \in B(\mathcal{H})} \left| \int_{\mathbb{R}^d} \mathcal{S}_\pi f d\mu \right|, \quad (1)$$

where $B(\mathcal{H})$ denotes the unit ball in $\mathcal{H}$. For any admissible kernel, this optimization problem is solved by $f^* = \frac{\psi}{\|\psi\|_{\mathcal{H}}}$ with $\psi = \int_{\mathbb{R}^d} \mathcal{A}_\pi^k d\mu$, where $\mathcal{A}_\pi^k(x) = \nabla \log \pi(x) \Phi_k(x) + \nabla \Phi_k(x) \in \mathcal{H}$, and Φ_k is the feature map associated with the kernel k . Consequently, the supremum evaluates to the RKHS norm of ψ , yielding the equivalent and broadly convenient characterization $\text{KSD}(\mu|\pi) = \|\psi\|_{\mathcal{H}}$.

2.2 Stein Variational Gradient Descent.

Given an arbitrary target distribution π and a reference distribution μ_0 , SVGD transforms μ_0 into an approximation of π by choosing $\mu_{n+1} := T_{\sharp} \mu_n$, where $T_{\sharp} \cdot$ is the push-forward operator for the map $T(x) = x + \gamma \psi^{\mu_n}(x)$, with the vector field $\psi^{\mu} = \int_{\mathbb{R}^d} \mathcal{A}_\pi^k d\mu$ for $\mu \in \mathcal{P}(\mathbb{R}^d)$ being the direction of steepest descent. This is motivated by the fact that the solution of Equation (1) implies that

$\frac{\psi^{\mu_n}}{\|\psi^{\mu_n}\|_{\mathcal{H}}}$ is the minimizer of $\frac{d}{d\gamma}\text{KL}(T_{\sharp}\mu_n \parallel \pi)\Big|_{\gamma=0}$ in the unit ball of $\mathcal{H}$ [cf. 18, Theorem 3.1], where $\text{KL}(\cdot \parallel \cdot)$ denotes the KL-divergence. In particular, for this choice we have

$$\frac{d}{d\gamma}\text{KL}\left((\text{Id} + \gamma\psi^{\mu_n})_{\sharp}\mu_n \parallel \pi\right)\Big|_{\gamma=0} = -\text{KSD}^2(\mu_n|\pi). \quad (2)$$

Iteratively applying the maps T generated in this way to a particle set $\{X_0^i\}_{i=1}^M$ sampled from μ_0 leads to the following particle updates:

$$X_{n+1}^i = X_n^i + \frac{\gamma}{M} \sum_{j=1}^M k(X_n^i, X_n^j) \nabla \log \pi(X_n^j) + \nabla_{X_n^j} k(X_n^i, X_n^j). \quad (3)$$

3 Adaptive kernel selection for SVGD

Because establishing convergence of SVGD with respect to the KL divergence is difficult and hinges on restrictive assumptions, a broad line of recent work has instead analyzed convergence in terms of the kernelized Stein discrepancy (KSD) [16, 33, 32]. Minimizing the KSD alone, however, does not by itself guarantee weak convergence: the sequence of mean-field measures may fail to be tight, a condition that is necessary for weak convergence of probability measures in Polish spaces. Moreover, as the optimization problem in Equation (1) makes clear, the RKHS structure—and hence the choice of kernel—directly determines how faithfully convergence in KSD translates into weak convergence [11].

To address this issue and strengthen convergence guarantees in a general way, we formulate our adaptive variant of SVGD over an arbitrary parameterized family of kernels $\{k_{\theta} \mid \theta \in \Theta\}$. For each $\theta \in \Theta$, we write KSD_{θ} for the corresponding *kernelized Stein discrepancy*, ψ_{θ}^{μ} for the optimal update direction, and Φ_{θ} for the associated feature map. This explicit and deliberately generic parameterization lets us adaptively select kernels throughout optimization, accounting for the kernel’s influence on convergence and lifting the limitations inherent in committing to a single fixed kernel.

Our approach rests on the principle that it is advantageous to select, at each stage, a kernel that maximizes the KSD between the empirical particle distribution and the target. The underlying intuition is general and straightforward: the instantaneous decrease in KL divergence under SVGD is proportional to the squared KSD at the current particle measure (see Equations (2) and (7)). Hence, at any iteration, choosing the kernel that yields the largest KSD amounts to maximizing the rate of KL decrease—a criterion that is meaningful regardless of the specific kernel family in use. This perspective is reinforced by geometric analyses of SVGD as a gradient flow, which show that kernels inducing larger KSD values enjoy more favorable convergence properties when the associated RKHSs are compared [28, 8].

Whereas vanilla SVGD typically seeks to guarantee convergence of the KSD for a single fixed kernel, our algorithm targets the worst-case KSD within the entire kernel class. Unlike the median heuristic used by Liu and Wang [18], the resulting procedure applies to any class of parameterized kernels: at each step, we seek an optimal parameter

$$\theta_n \in \operatorname{argmax}_{\theta \in \Theta} \text{KSD}_{\theta}(\mu_n|\pi).$$

We realize this idea by adjusting the kernel parameter with one or more gradient ascent steps on the KSD, using a step size $s > 0$, before executing the particle update; we also permit skipping the parameter update on some iterations to reduce runtime (see Algorithm 1). To enable this option, we introduce a user-specified decision variable `paramupdate`. We stress that the framework is modular: the base SVGD step for transporting the particles can be replaced by more advanced variants (e.g., adaptive step-size schedules, line searches, momentum, or second-order/preconditioned updates) without altering our kernel-selection mechanism, and the gradient ascent on the kernel parameter can likewise be swapped for alternative optimization schemes. This modularity makes the method broadly compatible with existing SVGD pipelines. Our implementation of the kernel update builds on the formula [6, 20]

$$\text{KSD}^2(\mu|\pi) = \int_{\mathbb{R}^d \times \mathbb{R}^d} u_\pi^k d(\mu \otimes \mu) \quad (4)$$

with

$$\begin{aligned} u_\pi^k(x, y) = & k(x, y) \nabla \log \pi(x) \cdot \nabla \log \pi(y) + \nabla \log \pi(y) \cdot \nabla_x k(x, y) \\ & + \nabla \log \pi(x) \cdot \nabla_y k(x, y) + \text{trace}(\nabla_x \nabla_y k(x, y)), \end{aligned} \quad (5)$$

from which the required gradient follows directly. Two remarks are in order regarding the additional computational cost of Ad-SVGd relative to vanilla SVGD, both of which help keep the method broadly practical:

- The gradient ascent steps for the kernel parameter reuse the same gradients $\nabla \log \pi(X_n^i)$ as the corresponding SVGD step, so Ad-SVGd requires no additional gradient evaluations of $\log \pi$ in any application.
- The associated overhead can be reduced further by using only a small number of update steps `nstepstheta` for the kernel parameter, or by updating the kernel parameter only once every few SVGD steps. When the number of particles M is large, one may approximate the KSD empirically from subsampled particles, keeping the approach scalable to large ensembles.

Algorithm 1: Ad-SVGD

Input: Initial particle set $\{X_0^i \in \mathbb{R}^d \mid i = 1, \dots, M\}$, kernel class $\{k_\theta \mid \theta \in \Theta\}$, initial kernel parameter $\theta_{-1} \in \Theta$, step sizes $\gamma, s > 0$, number of steps $nsteps, nstepstheta \in \mathbb{N}$

Output: Final particle set $\{X_{nsteps}^i \in \mathbb{R}^d \mid i = 1, \dots, M\}$

```
for  $n = 0$  to  $nsteps - 1$  do
  Decide  $paramupdate \in \{\text{True}, \text{False}\}$ ;
  if  $paramupdate$  then
     $\theta_n^0 \leftarrow \theta_{n-1}$ ;
    for  $\ell = 0$  to  $nstepstheta - 1$  do
       $\theta_n^{\ell+1} \leftarrow \theta_n^\ell + s \nabla_{\theta_n^\ell} \text{KSD}_{\theta_n^\ell}^2 \left( \frac{1}{M} \sum_{i=1}^M \delta_{X_n^i} \mid \pi \right)$ ;
    end
     $\theta_n \leftarrow \theta_n^{nstepstheta}$ ;
  else
     $\theta_n \leftarrow \theta_{n-1}$ ;
  end
  for  $i = 1$  to  $M$  do
     $X_{n+1}^i \leftarrow X_n^i + \frac{\gamma}{M} \sum_{j=1}^M k_{\theta_n}(X_n^i, X_n^j) \nabla \log \pi(X_n^j) + \nabla_{X_n^j} k_{\theta_n}(X_n^i, X_n^j)$ ;
  end
end
```

4 Convergence analysis

We consider a general target measure with Lebesgue density of the form $\pi(x) \propto \exp(-V)$ for a potential $V : \mathbb{R}^d \rightarrow \mathbb{R}$. To motivate our proposed Ad-SVGD, we show how recent convergence results for SVGD in the sense of KSD extend to the adaptive setting. More precisely, we generalize the convergence analysis of [32], which relies on Talagrand’s inequality—a condition that holds under mild assumptions on the target distribution and is weaker than the commonly employed logarithmic Sobolev inequality, thereby broadening the class of targets to which the analysis applies. We adopt the same assumptions as [32], now required to hold uniformly over all kernel parameters.

Assumption 1. *We assume that $V \in C^2$ such that $\int_{\mathbb{R}^d} \exp(-V(x)) dx < \infty$ and that the Hessian H of V is uniformly bounded w.r.t. the operator norm, i.e. there exists $L \geq 0$ such that $\|H(x)\|_{\text{op}} \leq L$ for all $x \in \mathbb{R}^d$.*

Assumption 2. *We assume that $\pi \in \mathcal{P}_1(\mathbb{R}^d)$ satisfies Talagrand’s inequality T1, which means that there exists $\lambda > 0$ such that*

$$\mathcal{W}_1(\mu, \pi) \leq \sqrt{2 \text{KL}(\mu \parallel \pi) / \lambda}$$

for all $\mu \in \mathcal{P}_1(\mathbb{R}^d)$.

Assumption 3. *We assume there exists $B > 0$ such that $\|\Phi_\theta(x)\|_{\mathcal{H}_0} \leq B$ for all $x \in \mathbb{R}^d$, $\theta \in \Theta$, and that $\nabla \Phi_\theta$ is continuous with $\|\nabla \Phi_\theta(x)\|_{\mathcal{H}}^2 = \sum_{i=1}^d \|\partial_i \Phi_\theta(x)\|_{\mathcal{H}_0}^2 \leq B^2$ for all $x \in \mathbb{R}^d$, $\theta \in \Theta$.*

4.1 Convergence analysis of SVGD under Talagrand's inequality

In the following, we denote

$$\mathcal{F}(\mu) := \text{KL}(\mu \parallel \pi).$$

and rely on the following fundamental inequality, which holds for any kernel parameter in the class. Given a fixed kernel parameter $\theta \in \Theta$ such that Assumption 3 is satisfied, define the pushforward measure

$$\tilde{\mu} = (I + \gamma g) \# \mu$$

for arbitrary $g \in \mathcal{H}$. Under Assumptions 1 and 3, let $\gamma, B > 0$, $\alpha > 1$ and $g \in \mathcal{H}_\theta$ such that $\gamma \|g\|_{\mathcal{H}_\theta} \leq \frac{\alpha-1}{\alpha B}$. Then, according to Proposition 3.1 in [32], it holds

$$\mathcal{F}(\tilde{\mu}) \leq \mathcal{F}(\mu) + \gamma \langle \psi_\theta^\mu, g \rangle_{\mathcal{H}_\theta} + \frac{\gamma^2 K}{2} \|g\|_{\mathcal{H}_\theta} \quad (6)$$

with $K = (\alpha^2 + L)B$. For the iterates (3) with a fixed kernel parameter $\theta \in \Theta$ one can then derive the descent condition [32, Theorem 3.2],

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2}\right) \text{KSD}_\theta^2(\mu_n | \pi), \quad (7)$$

provided that

$$\gamma \leq (\alpha - 1) \left(\alpha B (1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}}) \right)^{-1}. \quad (8)$$

The crucial step in establishing (7) is verifying $\gamma \|\psi_\theta^{\mu_n}\|_{\mathcal{H}_\theta} \leq \frac{\alpha-1}{\alpha B}$ via Talagrand's inequality, Assumption 2, which permits applying (6) with $g = \psi_\theta^{\mu_n}$. Condition (7) then implies that

$$\lim_{n \rightarrow \infty} \text{KSD}_\theta^2(\mu_n | \pi) = 0,$$

since $\sum_{n=0}^{\infty} \text{KSD}_\theta^2(\mu_n | \pi) \leq c_\gamma^{-1} \mathcal{F}(\mu_0)$ for $c_\gamma = \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2}\right) > 0$.

When introducing an adaptive kernel parameter choice (θ_n) , the inequality (7) changes to

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2}\right) \text{KSD}_{\theta_n}^2(\mu_n | \pi).$$

4.2 Convergence analysis of Ad-SVGd.

Suppose the adaptive SVGD iteration can be written in the following simplified form

$$\begin{aligned} \mu_{n+1} &= (I + \gamma \psi_{\theta_n}^{\mu_n}) \# \mu_n, \\ \theta_n &\in \operatorname{argmax}_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n | \pi), \end{aligned} \quad (9)$$

meaning that

$$\|\psi_{\theta_n}^{\mu_n}\|_{\mathcal{H}_{\theta_n}} = \max_{\theta \in \Theta} \text{KSD}_\theta(\mu_n | \pi).$$

We emphasize that this idealized formulation is available whenever the maximization of the KSD over the kernel parameter admits a (unique) closed-form solution. This is precisely the situation in

the multiple-kernel SVGD framework [1], which parameterizes the kernel as a convex combination of base kernels

$$k_\theta(x, y) = \sum_{i=1}^N \theta_i k_i(x, y) \quad \text{with} \quad \sum_{i=1}^N \theta_i = 1.$$

In the general case—which is our primary interest and covers arbitrary kernel families—the kernel parameter update θ_n in (9) must instead be approximated. For this general setup we assume throughout that $\operatorname{argmax}_{\theta \in \Theta} \operatorname{KSD}_\theta^2(\mu_n | \pi) \neq \emptyset$ for all $n \in \mathbb{N}$.

The following descent condition is a direct consequence of Theorem 3.2 in [32]:

Lemma 1. *Suppose that Assumptions 1-3 are satisfied. For any $\alpha > 1$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

there exists $c_\gamma > 0$ such that for all $n \in \mathbb{N}$

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - c_\gamma \max_{\theta \in \Theta} \operatorname{KSD}_\theta^2(\mu_n | \pi),$$

where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (9).

Corollary 2. *Under the same assumptions as in Lemma 1 it holds that $\lim_{n \rightarrow \infty} \max_{\theta \in \Theta} \operatorname{KSD}_\theta^2(\mu_n | \pi) = 0$.*

The situation changes when we assume that we can only approximately solve the maximization task of the KSD with respect to the kernel parameter. Suppose that $\max_{\theta \in \Theta} \operatorname{KSD}_\theta(\mu | \pi) < \infty$ for any $\mu \in \mathcal{P}(\mathbb{R}^d)$ and consider the alternating scheme

$$\begin{aligned} \theta_n &= \Psi_n(\theta_{n-1}, \mu_n), \\ \mu_{n+1} &= (\operatorname{Id} + \gamma \psi_{\theta_n}^{\mu_n})_\# \mu_n \end{aligned} \tag{10}$$

for some sequence of iterative update rules $\Psi_n : \Theta \times \mathcal{P}(\mathbb{R}^d) \rightarrow \Theta$ aiming to maximize $\operatorname{KSD}_\theta(\mu_n | \pi)$. In Algorithm 1, Ψ_n is the gradient ascent update for the KSD, but as noted above it may be replaced by any other suitable iterative optimization scheme without affecting the analysis. The only property we require of the update rule is the following convergence behavior.

Assumption 4. *We assume that there exists a sequence $(\varepsilon_n)_{n \in \mathbb{N}}$ such that $\lim_{n \rightarrow \infty} \varepsilon_n = 0$ and*

$$\max_{\theta \in \Theta} \operatorname{KSD}_\theta^2(\mu_n | \pi) - \operatorname{KSD}_{\theta_n}^2(\mu_n | \pi) \leq \varepsilon_n \quad \text{for all } n \in \mathbb{N}.$$

Using this assumption, we can make the following convergence guarantee.

Theorem 3. *Suppose that Assumptions 1-3 are satisfied. Under Assumption 4, for any $\alpha > 1$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

it holds that $\lim_{n \rightarrow \infty} \max_{\theta \in \Theta} \operatorname{KSD}_\theta(\mu_n | \pi) = 0$, where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (10).

Proof. Using Theorem 3.2 in [32] we obtain

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - c_\gamma \text{KSD}_{\theta_n}^2(\mu_n|\pi)$$

Since $(\mathcal{F}(\mu_n))_{n \in \mathbb{N}}$ is decreasing and bounded below by zero, it converges and, in particular, it is a Cauchy sequence. Hence, we have

$$c_\gamma \text{KSD}_{\theta_n}^2(\mu_n|\pi) \leq \mathcal{F}(\mu_n) - \mathcal{F}(\mu_{n+1}) \rightarrow 0$$

as $n \rightarrow \infty$. Under Assumption 4 we obtain

$$\max_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n|\pi) \leq \text{KSD}_{\theta_n}^2(\mu_n|\pi) + \varepsilon_n \rightarrow 0$$

as $n \rightarrow \infty$. □

Under the so-called Stein logarithmic Sobolev inequality, the convergence of Ad-SVGD can be strengthened further to convergence in the KL divergence. For several representative decay scenarios for ε_n we derive iteration complexity bounds for the mean-field limiting system (10), and this setting additionally lets us characterize explicitly the bias of Ad-SVGD when the KSD maximization is solved only to a fixed $\bar{\varepsilon}$ -accuracy at each iteration. The full details are deferred to Section B. Finally, we emphasize that it suffices for a *single* kernel in the class $\{k_\theta\}$ to satisfy the Stein logarithmic Sobolev inequality; this considerable flexibility makes the condition far easier to verify across problems than in the fixed-kernel case.

5 Numerical Experiments

In this section, we evaluate Ad-SVGD on a representative range of inference problems, beginning with two illustrative examples: a one-dimensional Gaussian mixture model and a linear inverse problem governed by an ordinary differential equation. To probe how the method behaves as problem dimensionality grows, additional experiments spanning increasing dimensions are provided in Section A. Finally, in Section 5.4 we demonstrate the advantages of Ad-SVGD over SVGD with the median heuristic on a Bayesian logistic regression model, a benchmark widely used throughout the SVGD literature, so as to situate our findings within an established evaluation setting.

5.1 Kernel parameterization

SVGD is most commonly deployed with kernels of the form

$$k_h(x, y) = \exp\left(-\frac{\|x - y\|_p^p}{h}\right),$$

where $\|\cdot\|_p$ denotes the p -norm on $\mathbb{R}^d$ [e.g. 18, 2, 8]. Our study centers on selection strategies for the parameter h , the kernel *bandwidth*, though the same adaptive principle carries over to other kernel parameters. The standard heuristic sets $h = \frac{\text{med}^p}{\log(M-1)}$, where med is the current median pairwise distance between particles; this choice aims to balance the two terms in the SVGD update (3) [18].

To exploit the flexibility of our adaptive method, we work with product kernels of the form

$$k_h(x, y) = \prod_{i=1}^d \exp\left(-\frac{|x_i - y_i|^p}{h_i}\right)$$

with parameter $h = (h_1, \dots, h_d)$, i.e. we allow a separate, dimension-dependent bandwidth per coordinate. The derivatives needed to apply Algorithm 1 with these kernels (see Equations (4) and (5)) are readily computed. For comparison, we also tested an adjusted median heuristic that takes a naive per-dimension median with these kernels; this variant, however, did not perform well and suffered from variance collapse.

Beyond the kernel families used in our experiments, the generality of Ad-SVGD readily accommodates more scalable kernel parameterizations for higher-dimensional problems. In particular, spectral or low-rank parameterizations can adapt to the dominant directions of variability without incurring the exponential cost of full product kernels, while Matérn or mixed-product constructions afford additional control over smoothness and anisotropy—illustrating that the method is not tied to any single choice of kernel.

5.2 Toy example

As a first, deliberately simple illustration, we consider the one-dimensional example from Liu and Wang [18]: a two-component Gaussian mixture $\pi = \frac{1}{3}\mathcal{N}(-2, 1) + \frac{2}{3}\mathcal{N}(2, 1)$. We use $M \in \{50, 200, 500\}$ particles, drawing the initial particle set from $\mathcal{N}(0, 1)$, and run SVGD for 10^4 steps with a step size of 1, employing kernels of the form described above with $p = 1$ across a range of fixed bandwidths h . To assess sample quality we adopt the Wasserstein 1-distance $\mathcal{W}_1$, computed via the explicit formula $\mathcal{W}_1(\mu, \nu) = \int_{\mathbb{R}} |F_\mu(x) - F_\nu(x)| dx$ (see Panaretos and Zemel [29]) together with an exact sample of size 10^5 . The results, reported in Figure 1, show that the algorithm performs well only for bandwidths within a narrow range. This pronounced sensitivity to h —a phenomenon that recurs well beyond this toy setting—underscores that a careful, adaptive selection strategy is essential in general.

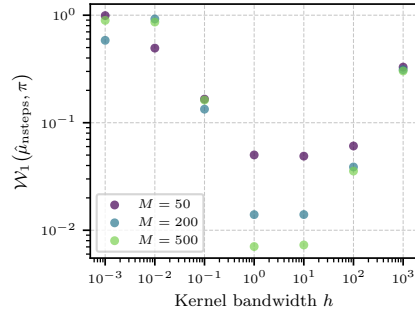


Figure 1: Final Wasserstein 1-distances for the one-dimensional example, obtained with SVGD across a range of fixed bandwidths h .

5.3 Linear inverse problem based on ODE

As a more realistic and representative setting, the following example—adapted from [38, Example 2.1]—situates Ad-SVGD within the broad class of Bayesian inverse problems governed by differential equations. We consider the one-dimensional differential equation

$$\begin{cases} -f''(s) + f(s) = u(s) & \text{for } s \in (0, 1) \\ f(s) = 0 & \text{for } s \in \{0, 1\} \end{cases} \quad (11)$$

and the associated inverse problem of recovering the right-hand side $u(\cdot) \in L^2([0, 1])$ from discrete noisy observation points of the solution $f \in H^2([0, 1]) \cap H_0^1([0, 1])$. These observations are described by

$$y = \Phi(u) + \varepsilon \in \mathbb{R}^{N_{\text{obs}}},$$

where $\varepsilon \in \mathbb{R}^{N_{\text{obs}}}$ is observational noise and the forward operator $\Phi : L^2([0, 1]) \rightarrow \mathbb{R}^{N_{\text{obs}}}$ is defined by $\mathcal{O} \circ H^{-1}$, with $H(f) = -f'' + f$ and $\mathcal{O}(f) = (f(s_1), \dots, f(s_{N_{\text{obs}}}))^\top \in \mathbb{R}^{N_{\text{obs}}}$ being the observation operator at N_{obs} equidistant points $s_k = \frac{k}{N_{\text{obs}}}$, $k = 1, \dots, N_{\text{obs}}$.

For the Bayesian formulation of the inverse problem, we consider a Gaussian process (GP) prior for u given by the truncated Karhunen-Loève (KL) expansion

$$u(\cdot, x) = Ax = \sum_{k=1}^{N_x} x_k \psi_k,$$

where $\psi_k(s) = \sqrt{2} \sin(\pi k s)$ and $x_k \sim \mathcal{N}(0, \lambda_k)$ independently with $\lambda_k = 50k^{-2}$.

The resulting problem is to estimate the KL coefficients $x = (x_1, \dots, x_{N_x})^\top \in \mathbb{R}^{N_x}$ with prior given by $\mathcal{N}(0, \Gamma_0)$ with $\Gamma_0 = \text{diag}(\lambda_1, \dots, \lambda_{N_x})$. Assuming $\varepsilon \sim \mathcal{N}(0, \Gamma)$, this leads to the posterior density

$$\pi(x) \propto \exp \left(-\frac{1}{2} \|\Gamma^{-1/2}(y - \Phi Ax)\|^2 - \frac{1}{2} \|\Gamma_0^{-1/2} x\|^2 \right), \quad x \in \mathbb{R}^{N_x}.$$

For the implementation, we replace H by a numerical discretization operator for Equation (11) on a grid with mesh size 2^{-8} and consider the fully observed system (i.e. $N_{\text{obs}} = 2^8$). We take $N_x = 16$ terms in the KL expansion of u and assume noise covariance $\Gamma = 10^{-3} \text{Id}_{N_{\text{obs}}}$. Reference observations $\bar{y} \in \mathbb{R}^{N_{\text{obs}}}$ are constructed by drawing $\bar{x} \sim \mathcal{N}(0, \Gamma_0)$ and setting $\bar{y} = \Phi A \bar{x}$.

We compare SVGD with the median heuristic (which we call Med-SVG) against our Ad-SVG across a range of particle ensemble sizes $M \in \{50, 100, 200\}$, using the kernels of Section 5.1 with $p = 1$. Both algorithms are run for $4 \cdot 10^5$ iterations with step size 10^{-3} for the particle updates and, following Liu and Wang [18], a variant of AdaGrad for adaptive step-size control. For Ad-SVG we use step size 10^{-5} for the bandwidth updates and refresh the bandwidth only once every 100 particle updates; under this schedule, we observe no meaningful runtime difference between Med-SVG and Ad-SVG, confirming that the added adaptivity comes essentially for free.

Figure 2 shows the GP reconstruction for a representative seed. Both methods approximate the mean well, but only Ad-SVG faithfully captures the posterior uncertainty. To quantify approximation quality more precisely, we use the Wasserstein 2-distance $\mathcal{W}_2(\mathcal{N}(\hat{\mu}, \hat{\Sigma}), \pi)$ between the posterior π and the normal distribution $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$, where $\hat{\mu}$ and $\hat{\Sigma}$ are the sample mean and covariance of the particle set [29, Section 3.2]. Since the target π is multivariate normal here, this admits the explicit formula

$$\mathcal{W}_2(\mathcal{N}(\hat{\mu}, \hat{\Sigma}), \pi)^2 = \|\hat{\mu} - \mu_\pi\|^2 + \text{trace} \left(\hat{\Sigma} + \Sigma_\pi - 2(\hat{\Sigma}^{1/2} \Sigma_\pi \hat{\Sigma}^{1/2})^{1/2} \right),$$

where μ_π and Σ_π are the mean and covariance of the posterior. We additionally compare the marginal variances of the final particle distribution against those of the posterior. Figure 3 reports these results aggregated over 56 random seeds (note that the posterior covariance does not depend on $\bar{y}$). Consistent with the picture that emerges across our experiments, Ad-SVG approximates the posterior more accurately than Med-SVG, which systematically underestimates

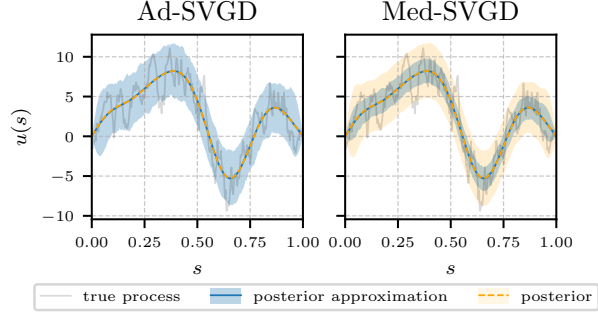


Figure 2: GP reconstruction for the ODE-based inverse problem, showing the mean and the 90% confidence interval.

interval.

Assuming $\varepsilon \sim \mathcal{N}(0, \Gamma)$, this leads to the posterior density

For the implementation, we replace H by a numerical discretization operator for Equation (11) on a grid with mesh size 2^{-8} and consider the fully observed system (i.e. $N_{\text{obs}} = 2^8$). We take $N_x = 16$ terms in the KL expansion of u and assume noise covariance $\Gamma = 10^{-3} \text{Id}_{N_{\text{obs}}}$. Reference observations $\bar{y} \in \mathbb{R}^{N_{\text{obs}}}$ are constructed by drawing $\bar{x} \sim \mathcal{N}(0, \Gamma_0)$ and setting $\bar{y} = \Phi A \bar{x}$.

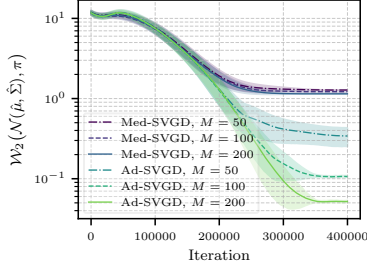
We compare SVGD with the median heuristic (which we call Med-SVG) against our Ad-SVG across a range of particle ensemble sizes $M \in \{50, 100, 200\}$, using the kernels of Section 5.1 with $p = 1$. Both algorithms are run for $4 \cdot 10^5$ iterations with step size 10^{-3} for the particle updates and, following Liu and Wang [18], a variant of AdaGrad for adaptive step-size control. For Ad-SVG we use step size 10^{-5} for the bandwidth updates and refresh the bandwidth only once every 100 particle updates; under this schedule, we observe no meaningful runtime difference between Med-SVG and Ad-SVG, confirming that the added adaptivity comes essentially for free.

Figure 2 shows the GP reconstruction for a representative seed. Both methods approximate the mean well, but only Ad-SVG faithfully captures the posterior uncertainty. To quantify approximation quality more precisely, we use the Wasserstein 2-distance $\mathcal{W}_2(\mathcal{N}(\hat{\mu}, \hat{\Sigma}), \pi)$ between the posterior π and the normal distribution $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$, where $\hat{\mu}$ and $\hat{\Sigma}$ are the sample mean and covariance of the particle set [29, Section 3.2]. Since the target π is multivariate normal here, this admits the explicit formula

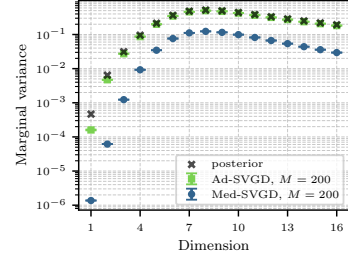
$$\mathcal{W}_2(\mathcal{N}(\hat{\mu}, \hat{\Sigma}), \pi)^2 = \|\hat{\mu} - \mu_\pi\|^2 + \text{trace} \left(\hat{\Sigma} + \Sigma_\pi - 2(\hat{\Sigma}^{1/2} \Sigma_\pi \hat{\Sigma}^{1/2})^{1/2} \right),$$

where μ_π and Σ_π are the mean and covariance of the posterior. We additionally compare the marginal variances of the final particle distribution against those of the posterior. Figure 3 reports these results aggregated over 56 random seeds (note that the posterior covariance does not depend on $\bar{y}$). Consistent with the picture that emerges across our experiments, Ad-SVG approximates the posterior more accurately than Med-SVG, which systematically underestimates

the uncertainty. Moreover, in contrast to Med-SVGD, the approximation quality of Ad-SVGD keeps improving as the number of particles grows beyond 50. Figure 4 depicts the behavior of the bandwidths selected by Ad-SVGD: the component-wise bandwidths stabilize more quickly than the approximation error does, and clear differences across components emerge, with the final bandwidths negatively correlated with the corresponding marginal variances.

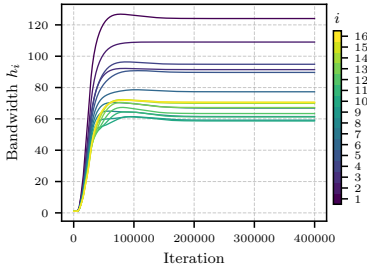


(a) Convergence of approximation error

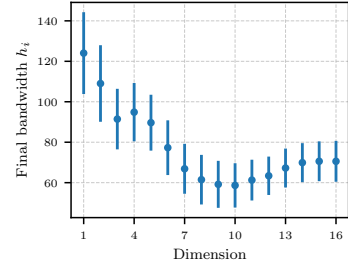


(b) Marginal variances

Figure 3: Aggregated results (mean and 95% confidence interval over 56 random seeds) for the ODE-based inverse problem, comparing Med-SVGD and Ad-SVGD.



(a) Evolution of bandwidth per dimension (mean)



(b) Final bandwidths (mean and 95% conf. interval)

Figure 4: Behavior of the bandwidth parameter for the ODE-based inverse problem under Ad-SVGD, aggregated over 56 random seeds.

5.4 Bayesian logistic regression

To evaluate Ad-SVGD on a standard real-data classification benchmark, we follow Liu and Wang [18] and Liu et al. [21] and consider a Bayesian logistic regression (BLR) model applied to the Covertypes data set [4]. For the regression weights w , we place a Gaussian prior $w | \alpha \sim \mathcal{N}(0, \alpha^{-1})$ with $\alpha \sim \text{Gamma}(1, 0.01)$ and seek the posterior of $x = [w, \alpha]$. Following Liu et al. [21], we use the No-U-Turn Sampler (NUTS) of Hoffman and Gelman [13] to generate a reference against which the posterior approximation quality of Med-SVGD and Ad-SVGD can be assessed. Again following Liu et al. [21], we evaluate the methods on subsets of the original data set of size 1000 and aggregate results over 10 random draws. For the classification task itself, we report the prediction accuracy over a test set obtained from the particle mean. As Figure 5 shows, Ad-SVGD outperforms Med-SVGD on nearly all seeds, attaining higher test accuracy while also staying closer to the MCMC refer-

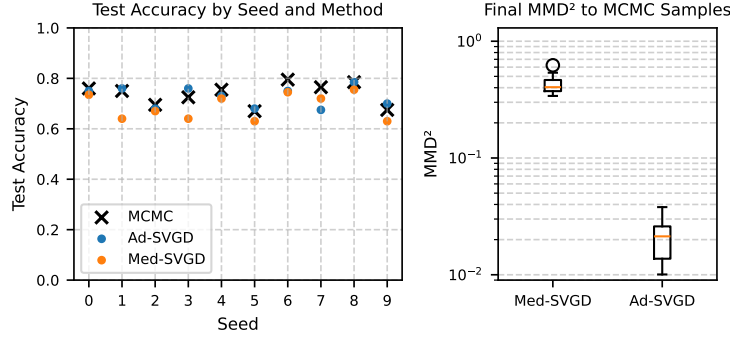


Figure 5: Approximation quality of the SVGD particles for Bayesian logistic regression, measured by prediction accuracy (left) and MMD^2 to the reference samples (right).

ence—evidence that the method’s benefits carry over from synthetic targets to a real classification problem.

To gauge the accuracy of the posterior approximation, the same figure also reports the squared maximum mean discrepancy (MMD^2) [25, Section 3.5] to the MCMC samples. The two methods differ markedly in how faithfully they capture the posterior. This is further confirmed by Figure 7, which displays the covariance matrices of the final particles alongside the MCMC reference, averaged over the 10 seeds (following the visualization style of Liu et al. [21]). Med-SVGd severely underestimates the posterior uncertainty, whereas Ad-SVGd yields a substantially better approximation. In addition, Figure 6 traces the evolution of the per-component bandwidths: starting from an uninformed initialization (equal to 1 in every component), Ad-SVGd recovers suitable bandwidths that differ from the initialization by several orders of magnitude, illustrating its ability to adapt automatically without problem-specific tuning.

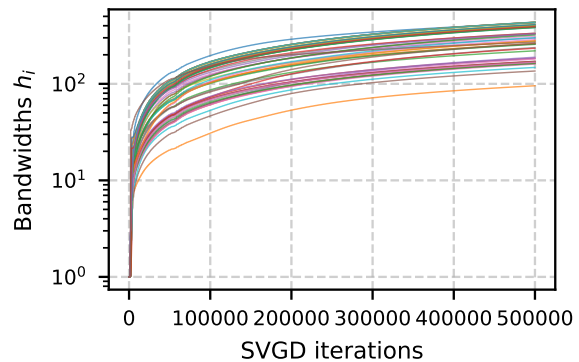


Figure 6: Bandwidth evolution for the BLR example.

6 Limitations and outlook

The principal limitation of our analysis is its reliance on Assumption 4. In ongoing work, we are investigating the general conditions under which Assumption 4 holds for the alternating gradient-ascent scheme of Algorithm 1 used in our experiments. While we do not yet provide guarantees that Assumption 4 is met, our implementation delivered consistently promising empirical results across the settings we studied.

Because our analysis targeted the original SVGd dynamics, and because the adaptive kernel-

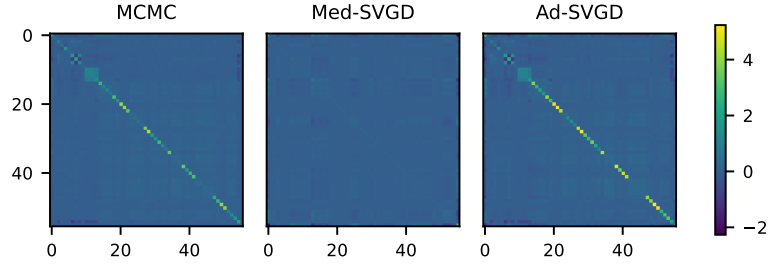


Figure 7: Covariance matrices of the final SVGD particle sets compared to the MCMC samples (mean over 10 seeds).

selection mechanism is deliberately agnostic to the underlying transport scheme, it would be natural and promising to combine it with recent variants such as sliced SVGD [10], Grassmann SVGD [21], or Stein transport [27]. More broadly, the framework opens the door to adaptive kernel learning across the wider family of Stein-based inference methods.

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A More numerical experiments

A.1 Gaussian mixture models

We revisit the example of Section 5.2. For the setup described there, Figure 9 shows histograms of the final particle distribution against the target density for $h \in \{0.001, 1, 1000\}$. We now turn to a direct comparison of Med-SVGD and Ad-SVGD across a range of particle counts M , comparing the final particle distributions after 10^4 iterations with step size 1.

Figure 8 reports the Wasserstein 1-distance between the final empirical particle distribution and the target

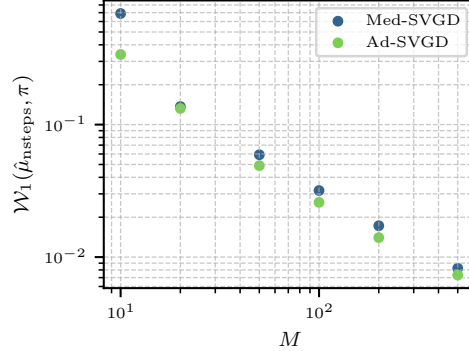


Figure 8: Approximation error for the 1D Gaussian mixture.

$M = 10, 20, 50, 100, 200, 500$. As expected, the approximation quality improves with N for both methods, and the two perform comparably here, both reaching a Wasserstein distance below 0.01 at $N = 500$.

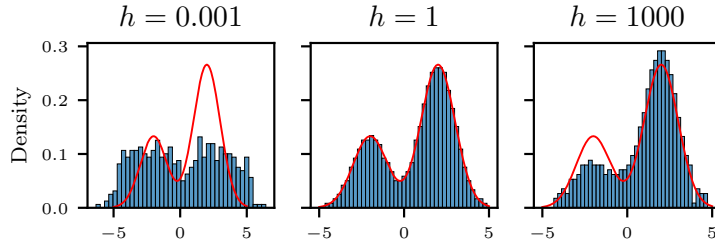


Figure 9: Histograms of the final particle set for fixed bandwidths $h = 0.001, 1, 1000$; the target density π is shown in red for comparison.

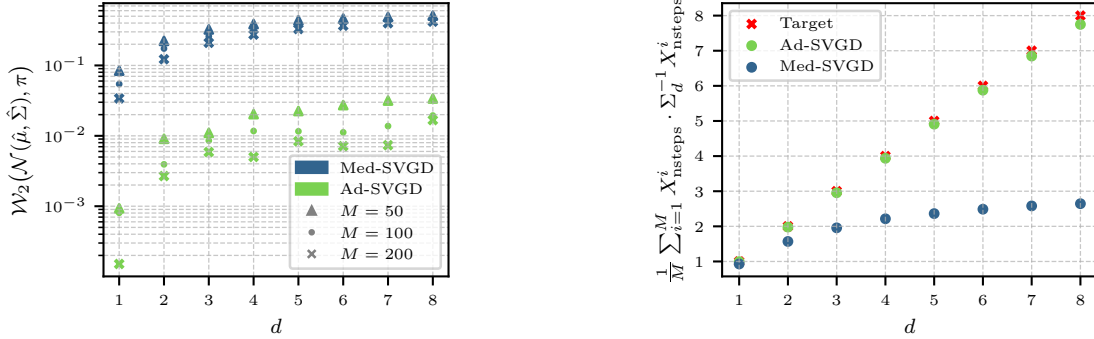
In this one-dimensional regime, both methods approximate the target well. Our adaptive bandwidth selection performs on par with the standard median heuristic, offering no clear advantage in this benign, low-dimensional scenario; as we show next, the picture changes substantially once the dimension grows.

A.2 Scaling dimension: multivariate normal distribution

To probe how the methods behave as dimensionality grows, we now consider the family of Gaussian targets $\pi_d = \mathcal{N}(0, \Sigma_d)$ with $\Sigma_d = \text{diag}(1, \frac{1}{2}, \dots, \frac{1}{d})$ for $d \in \{2, \dots, 8\}$. We used $\mathcal{N}(0, \frac{1}{d})^{\otimes d}$ as the initial particle distribution and ran the algorithms for 10^4 iterations with step size 0.1, reducing the step size and adjusting the number of iterations whenever needed for numerical stability.

We compare the approximation quality of the final particle sets produced by Med-SVGD and Ad-SVGD as the dimension increases. Because the Wasserstein 2-distance between two Gaussians

admits an explicit formula (see [29]), we take as our sample-quality measure the Wasserstein 2-distance between the target π_d and the Gaussian $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$, where $\hat{\mu}$ and $\hat{\Sigma}$ are the empirical mean and covariance of the particle set [29, Section 3.2]. Following the terminology of the Python Optimal Transport library [9] used in our computations, we call this the *Bures Wasserstein distance*. Figure 10a traces this measure for Med-SVGD and Ad-SVGD with $M = 50, 100, 200$ as the dimension grows. Across the entire range of d , Ad-SVGD substantially outperforms Med-SVGD, indicating that its advantage strengthens rather than fades with dimension.



(a) Approximation error for $M \in \{50, 100, 200\}$.

(b) χ^2 -test statistic for $M = 200$.

Figure 10: Results for multivariate normal targets of increasing dimension.

For the Bures Wasserstein distance, the smaller variances in the trailing components of the targets π_d contribute little; to expose the consequences of failing to capture the uncertainty in those components, we turn to the test statistic $X \cdot \Sigma_d^{-1} X$, which is χ^2 -distributed with d degrees of freedom (hence has expected value d) for $X \sim \mathcal{N}(0, \Sigma_d)$ (cf. [34, p. 27]). We evaluate the mean of this statistic over the particle set with $M = 200$ and compare it to the expected value d in Figure 10b. As soon as we move beyond one dimension, the statistic for Med-SVGD drifts progressively farther from the true expectation as d grows, whereas the statistic for Ad-SVGD remains close to it throughout. This lays bare the failure of Med-SVGD to approximate higher-dimensional targets and, conversely, the robustness of Ad-SVGD across the full range of dimensions. We give further detail on the final particle distributions for $M = 200$ and varying d in Tables 1 and 2, which list, for each d , the marginal variances of the particle set alongside those of the target (shown in the first row). Table 1 reports Med-SVGD and Table 2 reports Ad-SVGD. Ad-SVGD attains an accurate approximation of the target in every component, while Med-SVGD markedly underestimates the target’s uncertainty—consistent with the behavior already seen in Figure 10b. Finally, we zoom in on the marginal particle distributions for $M = 200$ and $d = 8$. For ease of visualization we normalized them by scaling each particle component by the inverse of the corresponding marginal standard deviation of the target (i.e. multiplying the i -th component by i), which renders each marginal of π_d standard normal. Figures 11 and 12 display histograms of these normalized marginals for $d = 8$ under Med-SVGD and Ad-SVGD, respectively, with a standard Gaussian density overlaid for reference. Once again, Ad-SVGD captures all marginals well while Med-SVGD underestimates the target’s uncertainty. The same pattern is evident in Figures 13 and 14, which compare the quantiles of the normalized marginal particle distributions against the target quantiles (a standard normal).

Table 1: Marginal variances of the final particle distribution across d -dimensional examples under Med-SVGD with $M = 200$; the marginal variances of the target π_d are shown for comparison.

component	1	2	3	4	5	6	7	8
Target	1.0000	0.2500	0.1111	0.0625	0.0400	0.0278	0.0204	0.0156
$d = 1$	0.9285							
$d = 2$	0.7921	0.1943						
$d = 3$	0.6803	0.1625	0.0697					
$d = 4$	0.6089	0.1440	0.0593	0.0311				
$d = 5$	0.5532	0.1275	0.0526	0.0271	0.0157			
$d = 6$	0.5190	0.1190	0.0481	0.0243	0.0140	0.0089		
$d = 7$	0.4900	0.1122	0.0449	0.0228	0.0131	0.0081	0.0052	
$d = 8$	0.4753	0.1077	0.0430	0.0215	0.0122	0.0074	0.0047	0.0032

Table 2: Marginal variances of the final particle distribution across d -dimensional examples under Ad-SVGD with $M = 200$; the marginal variances of the target π_d are shown for comparison.

component	1	2	3	4	5	6	7	8
Target	1.0000	0.2500	0.1111	0.0625	0.0400	0.0278	0.0204	0.0156
$d = 1$	0.9953							
$d = 2$	0.9907	0.2472						
$d = 3$	0.9867	0.2459	0.1095					
$d = 4$	0.9881	0.2467	0.1095	0.0610				
$d = 5$	0.9840	0.2433	0.1096	0.0616	0.0392			
$d = 6$	0.9858	0.2459	0.1090	0.0611	0.0392	0.0269		
$d = 7$	0.9856	0.2463	0.1086	0.0613	0.0390	0.0269	0.0199	
$d = 8$	0.9691	0.2409	0.1085	0.0611	0.0390	0.0268	0.0196	0.0150

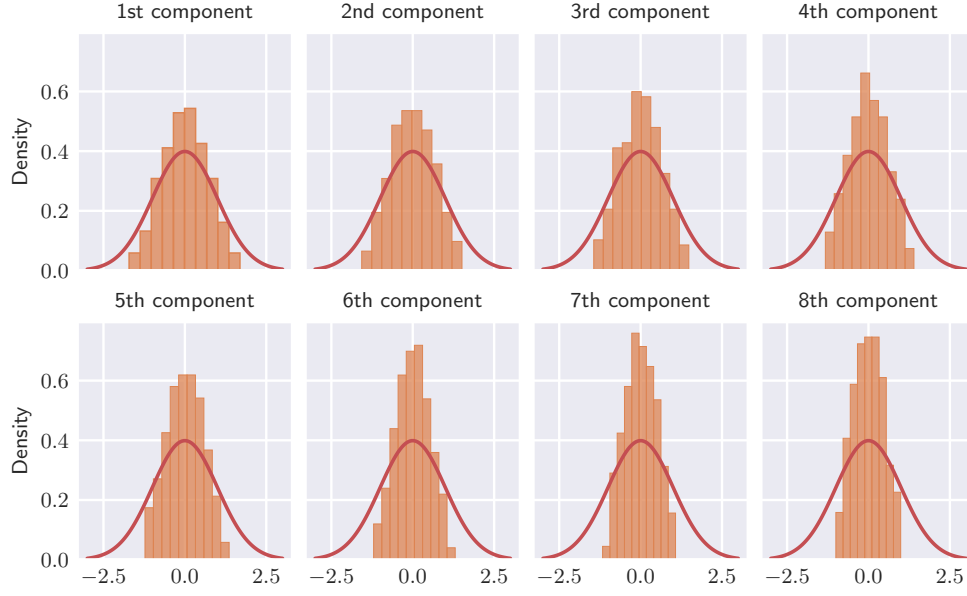


Figure 11: Component-wise histograms of the final particle set for the eight-dimensional example under Med-SVGD with $M = 200$, rescaled by the marginal target variances; a standard Gaussian density is shown for comparison.

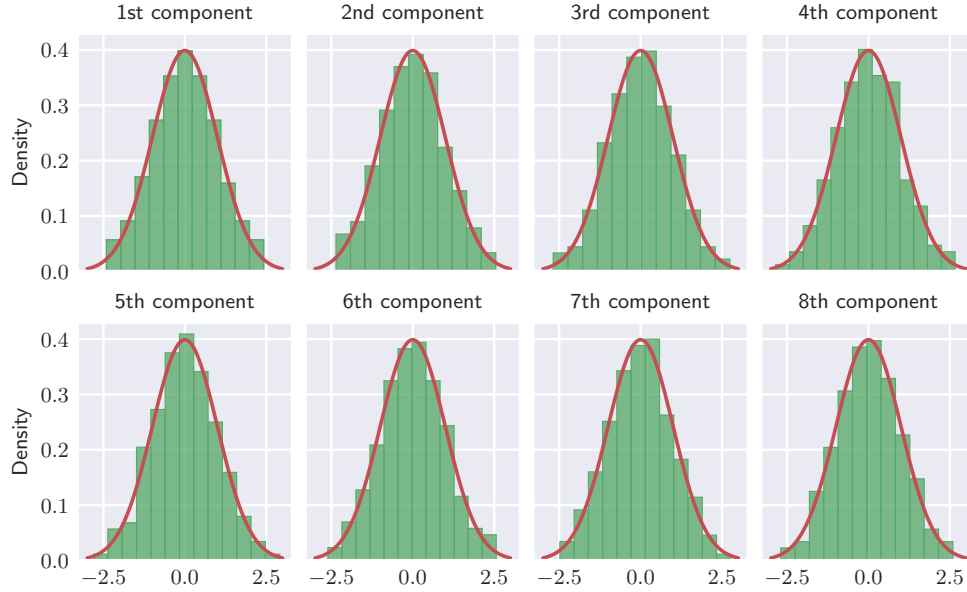


Figure 12: Component-wise histograms of the final particle set for the eight-dimensional example under Ad-SVGD with $M = 200$, rescaled by the marginal target variances; a standard Gaussian density is shown for comparison.

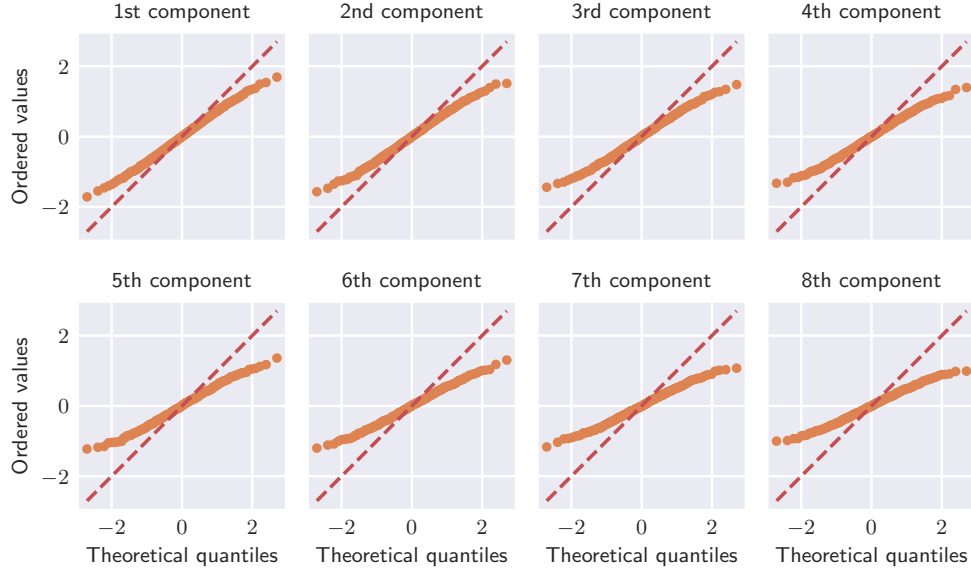


Figure 13: Q-Q plots comparing the marginals of the final particle distribution for the eight-dimensional example under Med-SVGd with $M = 200$, rescaled by the marginal target variances, against a standard normal distribution; the line of slope 1 through the origin is shown for comparison.

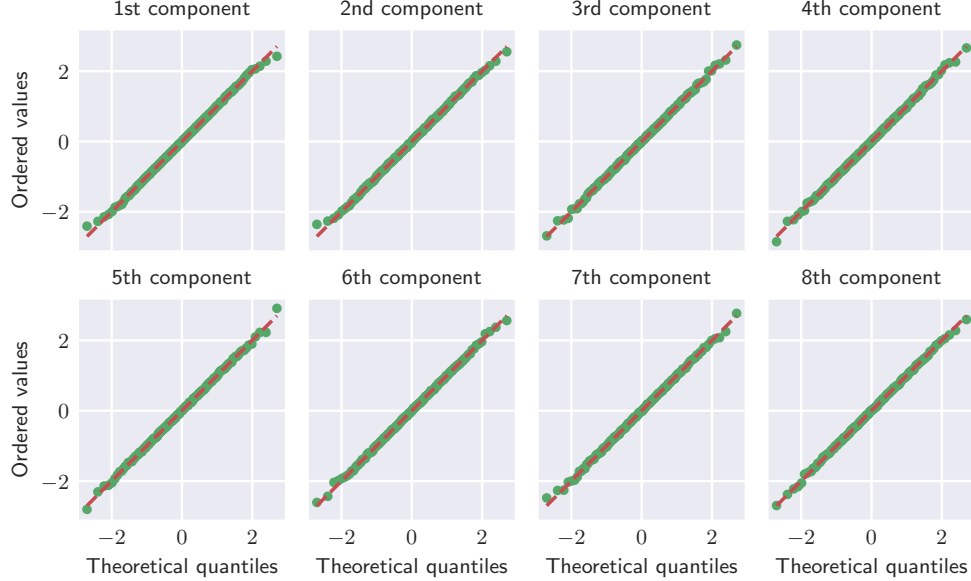


Figure 14: Q-Q plots comparing the marginals of the final particle distribution for the eight-dimensional example under Ad-SVGd with $M = 200$, rescaled by the marginal target variances, against a standard normal distribution; the line of slope 1 through the origin is shown for comparison.

A.3 Gaussian process inference

As a further, function-space example of scaling with dimension, we study the behavior of Ad-SVGD on the Gaussian process (GP) inference task proposed in [31]. We consider a GP on $[0, 1]$ represented by a truncated Karhunen-Loève (KL) expansion $u(s, x) = \sum_{k=1}^{N_x} x_k \psi_k(s)$ with basis functions $\psi_k(s) = \sqrt{2} \sin(k\pi s)$, where $x = (x_1, \dots, x_{N_x})^\top$ collects independent Gaussian random variables $x_k \sim \mathcal{N}(0, k^{-2})$. Observing the process at N_y equispaced points in $[0, 1]$, the task is to infer the coefficients x_k . For fixed N_x and N_y , this is an inverse problem with forward model $Y = AX + \varepsilon$, where the k -th column of $A \in \mathbb{R}^{N_y \times N_x}$ is $(\psi_k(s_1), \dots, \psi_k(s_{N_y}))^\top$ with $s_i = \frac{i}{N_y}$ for $i = 1, \dots, N_y$. The prior is $X \sim \mathcal{N}(0, \Sigma)$ with diagonal Σ having entries k^{-2} , $k = 1, \dots, N_x$, and we assume independent Gaussian noise $\varepsilon \sim \mathcal{N}(0, I_{N_y})$. Reference observations $\bar{y} \in \mathbb{R}^{N_y}$ are constructed by drawing $\bar{x} \sim \mathcal{N}(0, \Sigma)$ and setting $\bar{y} = A\bar{x}$.

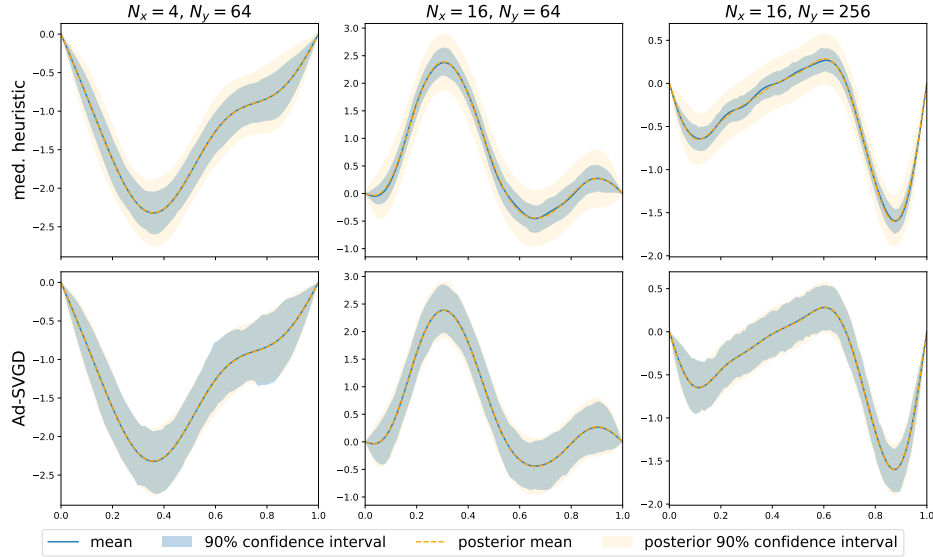


Figure 15: Estimated processes obtained with the median heuristic and Ad-SVGD using $M = 100$ particles across several model configurations, compared to the posterior.

We use SVGD to sample from the resulting posterior $X | Y = \bar{y}$ and compare the median heuristic against our adaptive approach. For SVGD with the median heuristic we use kernels of the form $k(x, y) = \exp(-\|x - y\|_1/h)$; for Ad-SVGD we use product kernels $k(x, y) = \prod_{i=1}^{N_x} \exp(-|x_i - y_i|/h_i)$ with parameter $h = (h_1, \dots, h_{N_x})$, i.e. a separate bandwidth per dimension. Following [18], we use a variant of Adagrad for step-size control in the scenarios with $N_x = 16$.

Figure 15 shows the resulting processes against the posterior for the configurations $N_x = 4$, $N_y = 64$; $N_x = 16$, $N_y = 64$; and $N_x = 16$, $N_y = 256$. As the dimension increases, SVGD with the median heuristic underestimates the posterior variance, whereas Ad-SVGD delivers a markedly better approximation—and this behavior holds consistently across the different numbers of GP observations, indicating that the advantage is not an artifact of a particular observation regime. Table 3 makes this quantitative by comparing the trace of the covariance of the SVGD particle distributions against that of the true posterior: the median heuristic severely underestimates the uncertainty, while Ad-SVGD captures the variance far more accurately.

N_x	4	8	16	16	16
N_y	64	64	64	128	256
theoretical	0.056	0.083	0.086	0.051	0.029
med. heuristic	0.026	0.023	0.022	0.012	0.006
Ad-SVGD	0.055	0.072	0.074	0.044	0.026

Table 3: Trace of the covariance of the final particle distribution ($M = 100$ particles) compared to the theoretical posterior across the different configurations, averaged over 25 runs.

B Refined theoretical analysis

Our convergence result can be strengthened further under a so-called Stein log-Sobolev inequality, which relates the KL divergence to the KSD. Duncan, Nüsken, and Szpruch [8] introduced this inequality and discussed conditions under which it may hold. Notably, the adaptive nature of our kernel selection lets us relax this requirement considerably: it suffices for a single kernel in the entire class to satisfy the log-Sobolev inequality, which broadens the range of targets for which the condition is applicable. More precisely, we make the following assumption.

Assumption 5. *Given a kernel class $\{k_\theta \mid \theta \in \Theta\}$, we assume that $\pi \in \mathcal{P}_1(\mathbb{R}^d)$ satisfies the generalized Stein log-Sobolev inequality with constant $\lambda > 0$:*

$$2\lambda \text{KL}(\mu \parallel \pi) \leq \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu \parallel \pi)$$

for all $\mu \in \mathcal{P}_1(\mathbb{R}^d)$.

This property lets us quantify convergence directly in terms of the KL divergence. We emphasize that, in this setting, an error bound on $\mathcal{F}(\mu_n)$ is available even under the weaker, approximate condition $\varepsilon_n \leq \bar{\varepsilon}$, without requiring $\varepsilon_n \rightarrow 0$ —a relaxation that widens the practical applicability of the guarantee.

Theorem 4. *Suppose that Assumptions 1 to 3 and 5 are satisfied. For any $\alpha > 1$ and $\gamma > 0$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

and

$$\rho := 1 - 2\lambda c_\gamma \in (0, 1) \quad \text{where} \quad c_\gamma := \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2} \right),$$

it holds that

$$\mathcal{F}(\mu_{n+1}) \leq \rho \mathcal{F}(\mu_n) + c_\gamma \varepsilon_n,$$

where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (10) and $\varepsilon_n \geq \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n \parallel \pi) - \text{KSD}_{\theta_n}^2(\mu_n \parallel \pi)$. In particular, if

1. $\varepsilon_n \leq \bar{\varepsilon}$ for all $n \in \mathbb{N}$, then

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + \frac{c_\gamma \bar{\varepsilon}}{1 - \rho}$$

for all $n \in \mathbb{N}$.

2. $\varepsilon \leq c_\varepsilon q^n$ for some $q \in (0, 1)$, $c_\varepsilon > 0$ and all $n \in \mathbb{N}$. Under $\rho \neq q$ we have

$$\mathcal{F}(\mu_n) \in \mathcal{O}(\max\{\rho, q\}^n)$$

and under $\rho = q$ we have

$$\mathcal{F}(\mu_n) \in \mathcal{O}(\tilde{q}^n)$$

for all $\rho < \tilde{q} < 1$.

3. $\varepsilon_n \leq \frac{c_\varepsilon}{(n+1)^p}$ for $c_\varepsilon, p > 0$ and all $n \in \mathbb{N}$, then

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + \frac{2^p c_\gamma c_\varepsilon}{n^p} \frac{1}{1 - \rho} + c_\gamma c_\varepsilon \frac{\rho^{\lfloor n/2 \rfloor}}{1 - \rho}$$

for all $n \in \mathbb{N}$.

Proof. Similar to the proof of Theorem 3, under Assumption 5 and the conditions on α, γ , we have

$$\begin{aligned} \mathcal{F}(\mu_{n+1}) &\leq \mathcal{F}(\mu_n) - c_\gamma \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n \mid \pi) + c_\gamma \varepsilon_n \\ &\leq \rho \mathcal{F}(\mu_n) + c_\gamma \varepsilon_n, \end{aligned}$$

where $c_\gamma = \gamma(1 - \frac{\gamma B(\alpha^2 + L)}{2}) > 0$. Iterating this recursion over $n \in \mathbb{N}$, we deduce the general error bound

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + c_\gamma \sum_{k=0}^{n-1} \rho^{n-1-k} \varepsilon_k.$$

We consider the three different cases separately:

1. Under $\varepsilon_n \leq \bar{\varepsilon}$, the first claim follows by the computation of the geometric series

$$\sum_{k=0}^{n-1} \rho^{n-1-k} \leq \sum_{k=0}^{\infty} \rho^k = \frac{1}{1 - \rho}.$$

2. Assuming that $\varepsilon_n \leq c_\varepsilon q^n$ for some $q \in (0, 1)$ and $c_\varepsilon > 0$ we have

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + c_\gamma c_\varepsilon \sum_{k=0}^{n-1} \rho^{n-1-k} q^k.$$

Suppose that $q > \rho$, then we have

$$\sum_{k=0}^{n-1} \rho^{n-1-k} q^k = \sum_{k=0}^{n-1} q^{n-1-k} \rho^k = q^{n-1} \sum_{k=0}^{n-1} (\rho/q)^k \leq q^{n-1} \frac{1}{1 - \rho/q} = q^n \frac{1}{q - \rho}$$

where we have used the formula for geometric series. Similarly, in the case of $q < \rho$, we can directly bound

$$\sum_{k=0}^{n-1} \rho^{n-1-k} q^k = \rho^{n-1} \sum_{k=0}^{n-1} (q/\rho)^k \leq \rho^{n-1} \frac{1}{1 - q/\rho} = \rho^n \frac{1}{(\rho - q)}.$$

Finally, in the case $q = \rho$, we simply bound $q^k < \tilde{q}^k$ for any $q < \tilde{q} < 1$ and deduce the claim line by line as before.

3. In the setting of $\varepsilon_n \leq \frac{c_\varepsilon}{(n+1)^p}$, we rewrite the upper bound on $\mathcal{F}(\mu_n)$ as

$$\begin{aligned}
\mathcal{F}(\mu_n) &\leq \rho^n \mathcal{F}(\mu_0) + c_\gamma \sum_{k=0}^{n-1} \rho^k \varepsilon_{n-1-k} \\
&= \rho^n \mathcal{F}(\mu_0) + c_\gamma \left(\sum_{k=0}^{\lfloor n/2 \rfloor} \rho^k \varepsilon_{n-1-k} + \sum_{k=\lfloor n/2 \rfloor + 1}^{n-1} \rho^k \varepsilon_{n-1-k} \right) \\
&\leq \rho^n \mathcal{F}(\mu_0) + \frac{2^p c_\gamma c_\varepsilon}{n^p} \frac{1}{1-\rho} + c_\gamma c_\varepsilon \frac{\rho^{\lfloor n/2 \rfloor}}{1-\rho}.
\end{aligned}$$

Here, we have used $\varepsilon_{n-1-k} \leq \frac{c_\varepsilon}{(n-k)^p} \leq \frac{2^p c_\varepsilon}{n^p}$ for all $k \leq \lfloor n/2 \rfloor$ and $\varepsilon_k \leq c_\varepsilon$ for all $k \in \mathbb{N}$.

□